

Ü 4

$$f(x) = (2-x)e^x$$

$$f'(x) = (-1)(e^x) + (2-x)(e^x) = -e^x + 2e^x - xe^x = e^x - xe^x = e^x(1-x)$$

$$f'(x) > 0 \Rightarrow e^x(1-x) > 0 \Rightarrow 1-x > 0 \Rightarrow -x > -1 \Rightarrow x < 1$$

$$\max f(x) = f(1) = e$$

$$f(-1) = (2+1)e^{-1} = 3 \cdot \frac{1}{e} = \frac{3}{e}$$

$$f(2) = (2-2)e^2 = (0)e^2 = 0$$

$$\min f(x) = f(2) = 0$$

Ü 5

$$f(x) = x - \sqrt{x^2 - 4} \quad x^2 - 4 \geq 0$$

$$\lim_{x \rightarrow -\infty} x - \sqrt{x^2 - 4} = -\infty$$

$$\lim_{x \rightarrow -\infty} \frac{f(x)}{x} = \frac{x - \sqrt{x^2 - 4}}{x} = 1 + \sqrt{1 - \frac{4}{x^2}} = 2 = m$$

$$\lim_{x \rightarrow -\infty} f(x) - m = f(x) - 2x = x - \sqrt{x^2 - 4} - 2x = -x - \sqrt{x^2 - 4} \quad [+ \infty - \infty]$$

$$\lim_{x \rightarrow -\infty} \frac{-x - \sqrt{x^2 - 4}}{-x + \sqrt{x^2 - 4}} = \frac{x^2 - (x^2 - 4)}{-x + \sqrt{x^2 - 4}} = \frac{4}{-x + \sqrt{x^2 - 4}} = 0 \quad y = 2x$$

$$\lim_{x \rightarrow +\infty} x - \sqrt{x^2 - 4} = +\infty - \infty$$

$$\lim_{x \rightarrow +\infty} \frac{(x - \sqrt{x^2 - 4})(x + \sqrt{x^2 - 4})}{(x + \sqrt{x^2 - 4})} = \frac{x^2 - (x^2 - 4)}{x + \sqrt{x^2 - 4}} = \frac{4}{x + \sqrt{x^2 - 4}} = 0 \quad \text{asintoto orizzontale}$$

Esercizio 6

$$f(x) = \begin{cases} a_1 = 1 \\ a_{n+1} = \frac{a_n^2 + 4}{2a_n} \quad \text{per } n \geq 1 \end{cases}$$

$$1) a_{n+1} = f(a_n)$$

$$f(x) = \frac{x^2 + 4}{2x} \quad a_{n+1} = f(a_n) \quad \forall n \geq 1$$

Studio $f(x)$

$$\text{Dominio} = \mathbb{R} \setminus \{0\}$$

$$f(-x) = \frac{-x^2 + 4}{-2x} = \frac{x^2 + 4}{2x} = -f(x) = \text{Dispari}$$

Segno

$$f(x) > 0$$

$$\frac{x^2 + 4}{2x} > 0 = \begin{cases} x^2 + 4 > 0 \\ 2x > 0 \end{cases} \quad \forall x \quad x > 0$$

Limiti

$$\lim_{x \rightarrow -\infty} \frac{x^2 + 4}{2x} = -\infty \quad \lim_{x \rightarrow 0^-} \frac{x^2 + 4}{2x} = -\infty \quad \lim_{x \rightarrow 0^+} \frac{x^2 + 4}{2x} = +\infty$$

$$f'(x) > 0 \Rightarrow x < -2 \vee x > 2$$

$x = -2$ massimo

$x = 2$ minimo

(ii) Punti fissi $f(x) = x$

$$\frac{x^2 + 4}{2x} = x$$

$$x^2 + 4 = 2x^2$$

$$-x^2 + 4 = 0$$

$$x^2 = 4 \quad x = \pm 2$$

3) $a_n \in [2, 5/2]$ per $n \geq 2$

$$d_n = |a_n - 2| \leq 1/2 \quad \text{per } n \geq 2$$